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OOP

Final Project

Part 1 & 2

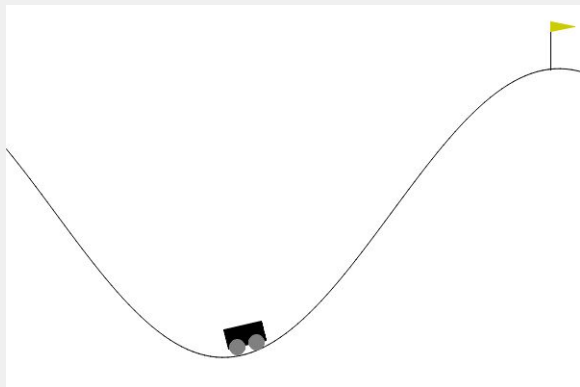
Part 1: We successfully installed and ran

Part 2: We tuned the parameters to:

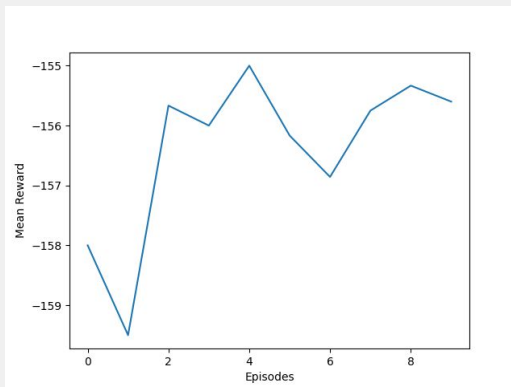
$$Q(s, a) \leftarrow Q(s, a) + \underbrace{0.034755}_{\text{learning rate}} [r + \underbrace{0.983}_{\text{discount factor}} \max_{a'} Q(s', a') - Q(s, a)]$$

$$\epsilon = 1$$

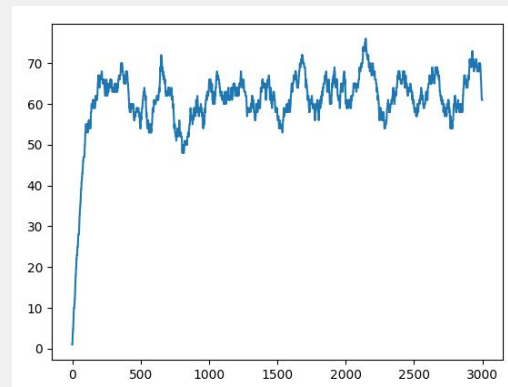
$$\epsilon \text{ decay rate} = 0.000096$$



Mountain Car



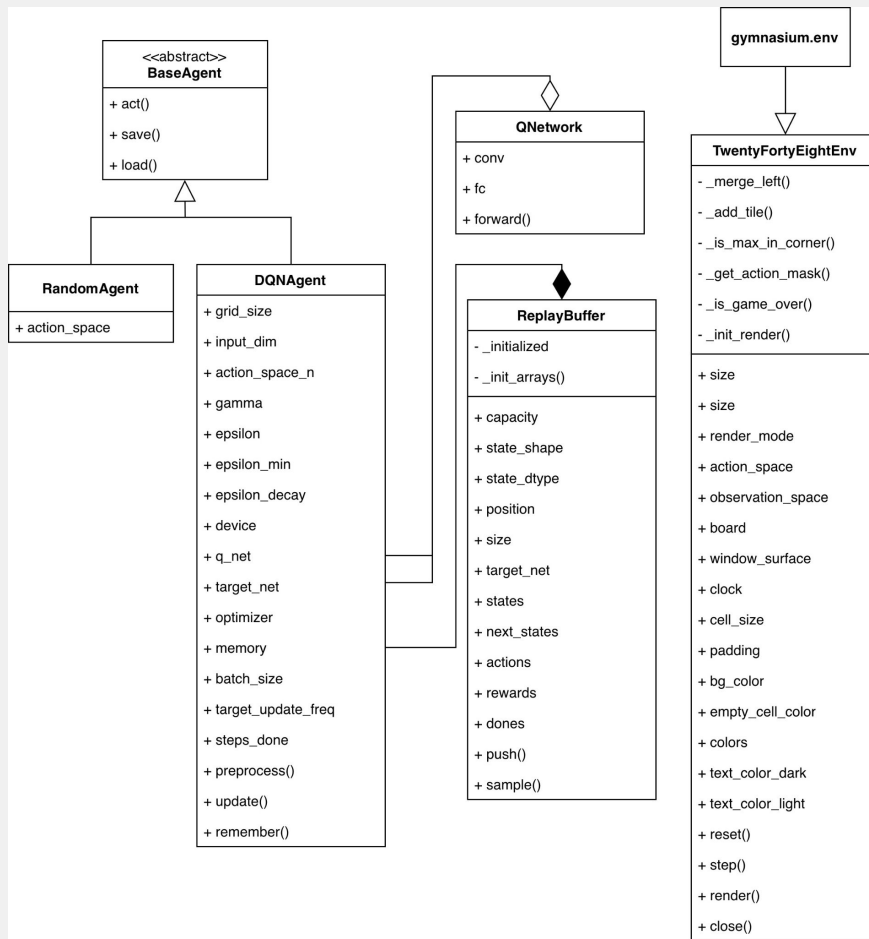
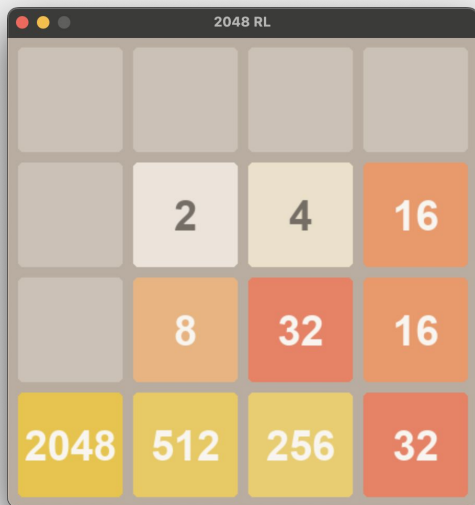
Training Score per episode (pt.1)

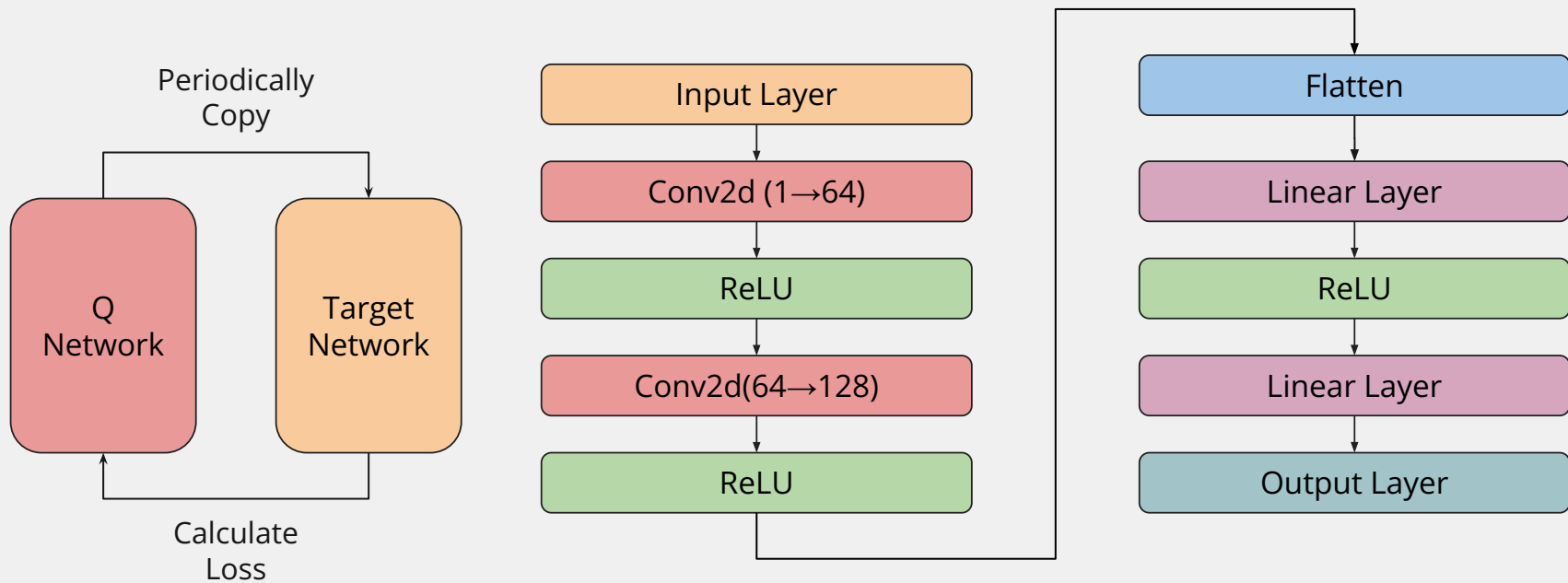


Training Score per episode (pt.2)

Part 3

Goal: Build an agent that runs 2048 by itself

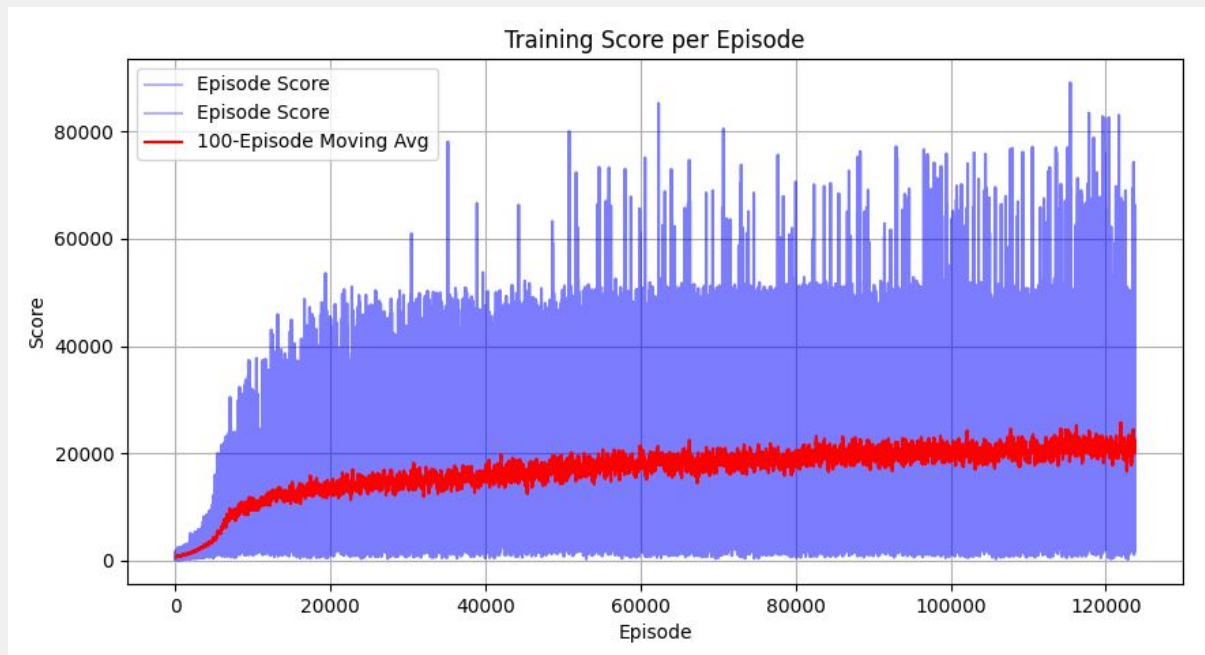




Part 3

Implementation: Deep Q-Network

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Testing for 10 episodes...
Using device: mps
Test Episode 1, Score: 20792.0, Max Tile: 1024
Test Episode 2, Score: 8508.0, Max Tile: 256
Test Episode 3, Score: 8228.0, Max Tile: 512
Test Episode 4, Score: 34736.0, Max Tile: 2048
Test Episode 5, Score: 24168.0, Max Tile: 1024
Test Episode 6, Score: 39128.0, Max Tile: 2048
Test Episode 7, Score: 39852.0, Max Tile: 2048
Test Episode 8, Score: 9464.0, Max Tile: 512
Test Episode 9, Score: 39484.0, Max Tile: 2048
Test Episode 10, Score: 36348.0, Max Tile: 2048
```



Part 3

*Thank
you!*